

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 02



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Recap of Previous Lecture



Topic

Calendar

ODD DAYS



Topics to be Covered



Topic-1

Clock



[MCQ]



#Q. The GATE 2020 result was declared on March 16, 2020. What day of the week was it?

- A** Saturday
- B** Sunday
- C** Monday
- D** None

2000 $\rightarrow$ 0

19 $\rightarrow$ 2

L.Y. 4
N.Y. 15

8 15

① + ① = 2

2019 $\rightarrow$ 2

J $\rightarrow$ 3
F $\rightarrow$ 1
M $\rightarrow$ 2

6

04
16

x4

2+6 = 8 $\rightarrow$ odd day

7

[MCQ]



#Q. Mahatma Gandhi was born on 2nd October 1869. What was the week Day?

SATURDAY

1868

$4 + 2 = 6$

Assignment

A

Monday

B

Thursday

C

Tuesday

D

Saturday

$1800 \rightarrow 30$

$68 \rightarrow 10$

$L-4$ $N-4$

6 54 51 2

$J-3$ $J-2$
 $F-0$ $J-3$
 $M-3$ $A-3$
 $A-2$ $S-2$
 $R-3$ $O-2$
 2

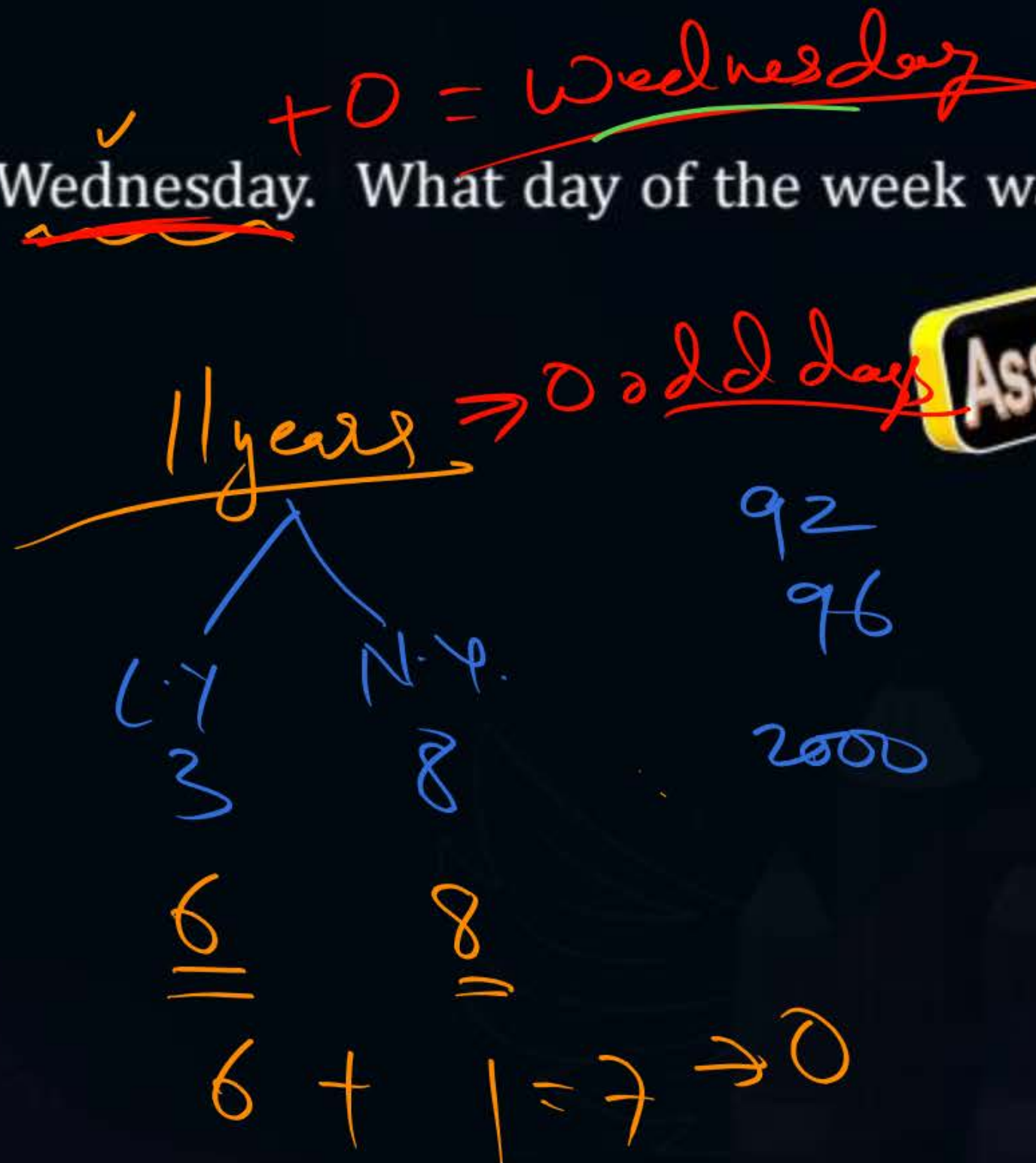
17 68

[MCQ]



#Q. If January 1st, 1992 was a Wednesday. What day of the week was January 1st, 2003?

- A** Sunday
- B** Wednesday
- C** Thursday
- D** Friday



Assignment

92
96
2000

CALENDAR REPETITION

2017 → 1

2018 → 1

2019 → 1

2020 → 2

2021 → 1

→ 2022

2011 → 1

2012 → 2

2013 → 1

2014 → 1

2015 → 1

2016 → 2

2022

leap years repeats only after 28 years
 $2016 + 28 \Rightarrow 2044$

leap year repeats after 28 years
 normal year repeats after 6 years
 normal year repeats after 11 years
 normal year repeats after 11 years
 leap year repeats after 28 years

·
·
·

normal year should repeat at normal year and leap year should repeat at leap year only along with $\%7 == 0$ in

[MCQ]

#Q. 2005 calendar is same as which of the given years?

N.Y.

+6

L.Y.

A 2009

B 2016

C 2011

D 2015

N.Y.

2005 → 1

2006 → 1

2007 → 1

2008 → 2

2009 → 1

2010 → 1

2011

[MCQ]



#Q. When does 2016 calendar repeats?

+28

2044

$$\begin{array}{r} 2013 \\ +6 \\ \hline 2019 \end{array}$$

L.Y. $\rightarrow$ 28 yrs

N.Y. $\rightarrow$ 6 yrs

N.Y. $\rightarrow$ 11 yrs

N.Y. $\rightarrow$ 11 yrs

L.Y. $\rightarrow$ 28 yrs

A 2027

B 2040

C 2021

D 2044

$$\begin{array}{r} 2012 \\ +28 \\ \hline 2040 \end{array}$$

$$\begin{array}{r} 2004 \\ +28 \\ \hline 2032 \end{array}$$

Conditional Question

Q. If a year consist of 302 days

5) $302 \div 60$ and every week has 5 days, then how many ODD DAYS that year has?

300

2

odd days

Conditional Question

✓
Tue Thurs
 Sat ✓

Q. The last day of a century CANNOT
be which day?

100 → 5 Fri
200 → 3 Wed
300 → 1 Mon
400 → 0 Sun

[MCQ]



#Q. The day of the March 16th of any year is the same day of the week as the corresponding date in which month of the same year?

- A** July
- B** November
- C** April
- D** September

March → 3
April → 2
May → 3
June → 2
July → 3

Aug → 3
Sep → 2
Oct → 3
→ Nov

#Q. A person was born on fifth Monday of February in a particular year.
Which one of the following statements is correct based on the above information?

1st Feb $\rightarrow$ Mon

29th Feb $\rightarrow$ Mon

A

The 2nd February of that year is a Tuesday.

B

There will be five Sundays in the month of February in that year.

C

The 1st February of that year is a Sunday.

D

All Mondays of February in that year have even dates.

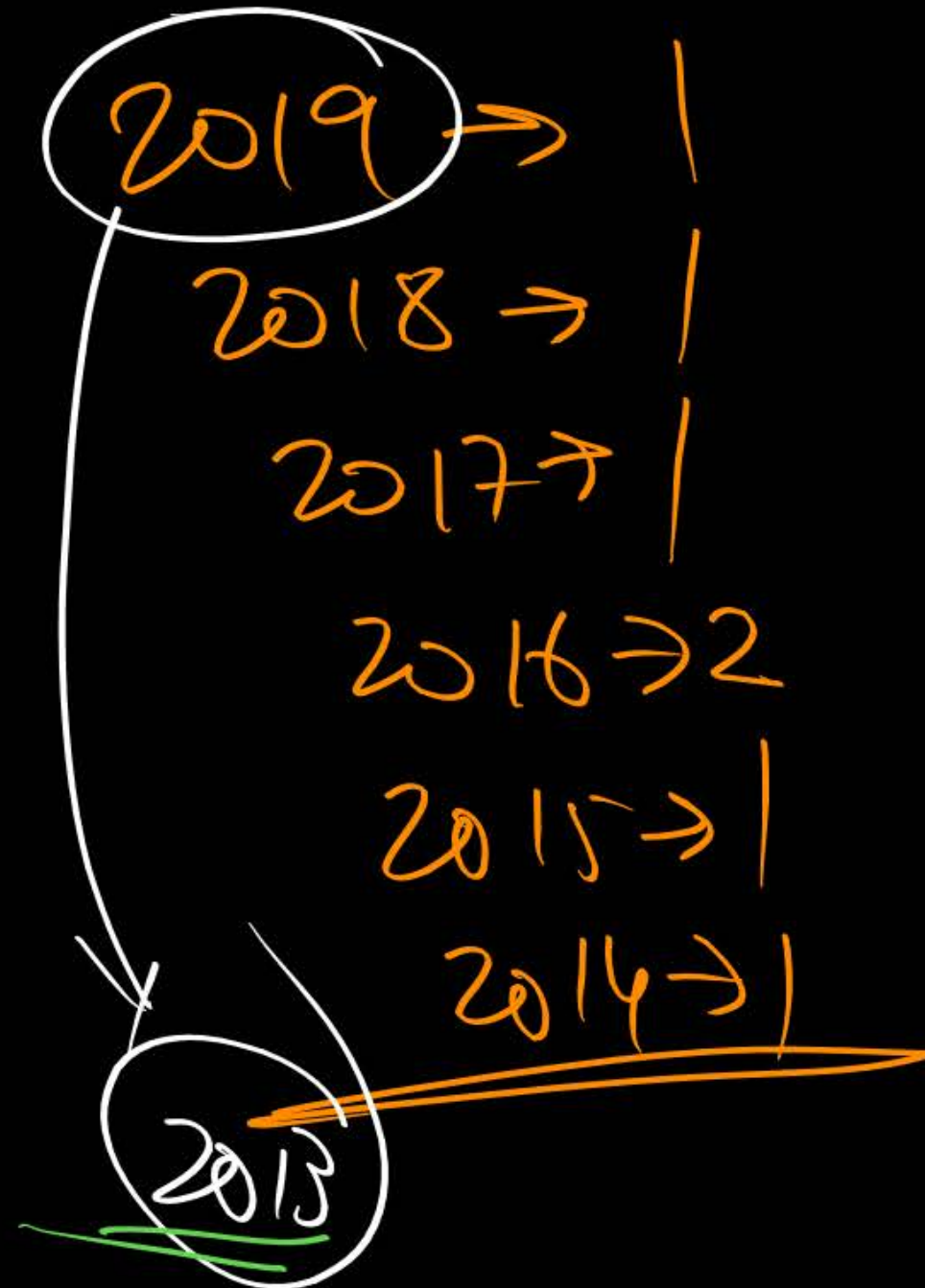
#Q. For the year 2019, which of the previous year's calendar can be used?

A 2011

B 2012

C 2013

D 2014



CLOCK



Relative Speed:

opposite $\Rightarrow (+)$

Same $\Rightarrow (-)$

Relative Speed:

1. Hands

Minimum

→ 2

2. Movement

Same

3. Relative Speed



(H.H.)
Hour hand | (M.H.)
Minute hand

Dial of Clock

M.H.

$$= \frac{360^\circ}{12}$$

6° / minute



$$\frac{360^\circ}{12} = 30^\circ$$

$$\underline{\underline{H.H.}} = \frac{360^\circ}{60}$$

$$= \frac{1^\circ}{2} \text{ minute}$$

$$= \underline{\underline{0.5^\circ / \text{minute}}}$$

To be Noted:

✓ Minute Hand Covers 30° in 5 minutes $30^\circ/5 =$

? $6^\circ/\text{min.}$

✓ Hour Hand Covers 30° in 60 minutes $30^\circ/60$

$= \frac{1}{2}^\circ / \text{min}$
or

$0.5^\circ/\text{min}$

Relative Speed

$5.5^\circ/\text{minute}$

Different Patterns of Questions:

- ① Time $\xrightarrow{v}$ Angle?
- ② Angle $\xrightarrow{v}$ Time?
- ③ Gain OR losing

First Pattern:

Time $\rightarrow$ Angle?

2:00
 60°



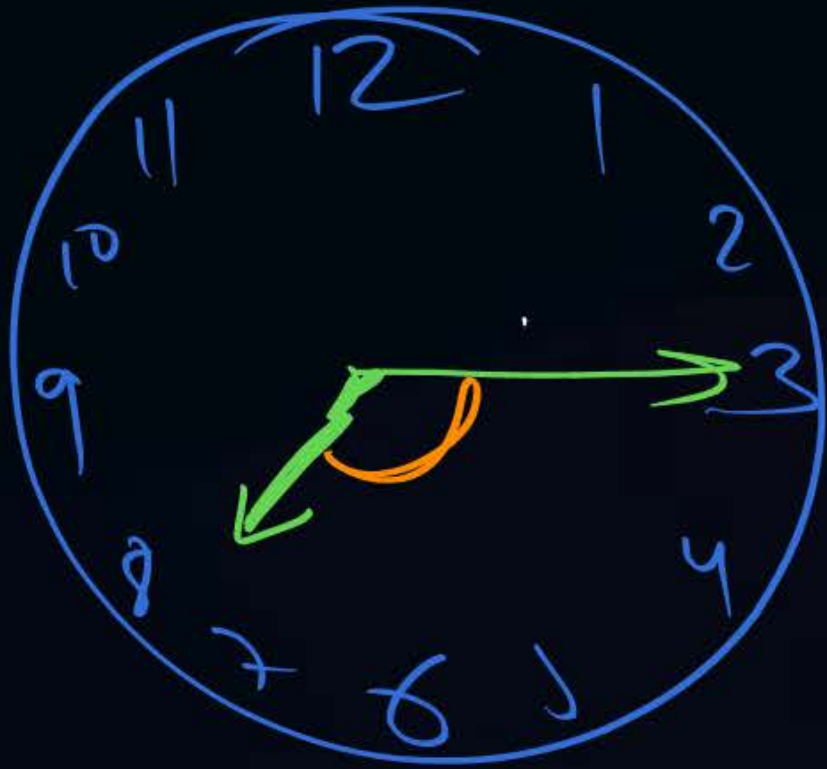
4:00

9:00
 270°

7:00

210°

Random time given:



9:23

143.5°

$9 \rightarrow 270^\circ$

$23 \times 5.5 \rightarrow 126.5^\circ$

143.5°

7:15

227.5°

$7 \rightarrow 210^\circ$

$15 \times 5.5 \rightarrow 82.5^\circ$

127.5°

Questionnaire:

5:30 → 15°

360-15

345°

5 → 150°

30 × 5.5 → 165°

15°



Questionnaire:

352.5° w

3:15 → 7.5° ✓

3 → 90°

15 × 5.5 → 82.5°

7.5°



Questionnaire:

260°

$4:40 \rightarrow 100^\circ$

$4 \Rightarrow 120^\circ$

$40 \times 5.5 \Rightarrow 220^\circ$

100°



[MCQ]



Assignment

#Q. At 9:45, the two hands of a clock make an angle of?

- A** 60°
- B** 45°
- C** $33 \frac{1}{3}^\circ$
- D** $22 \frac{1}{2}^\circ$

Two Answers:

9:45



$360 - 22.5^\circ$

347.5



2 mins Summary



Topic

Clock

S.S°/minute

Time [✓] → Angle?



THANK - YOU